

EXERCISE 7: UNION & UNION ALL

UNION : Removes duplicates

UNION ALL : Does not remove duplicates.

Table: online sales

sale_id	customer_name	amount	sale_date
1	Alice	150	2025-01-12
2	Brian	250	2025-02-05
3	Carl	300	2025-03-10
4	Daniel	220	2025-04-15
5	Emma	180	2025-05-07

Table: store sales

sales_id	customer_name	amount	sale_date
1	Fiona	200	2025-01-20
2	Brian	250	2025-02-08
3	George	310	2025-03-25
4	Alice	150	2025-04-18
5	Henry	270	2025-05-05

① Unique customer Names

List all unique customer names from both tables

using UNION

→ customer_name

→ SELECT customer - name
FROM online - sales

UNION

SELECT customer - name
FROM store - sales;

customer - name
Alice
Brian
Carol
Daniel
Emma
Fiona
George
Henry

② All customers (including duplicates)

List all customer names from both tables using

UNION ALL

→ SELECT customer - name

FROM online - sales

UNION ALL

SELECT customer - name

FROM store - sales;

customer - name
Alice
Brian
Carol
Daniel
Emma
Fiona
Brian
George
Alice
Henry

3. Unique Sale Dates

Show all unique sale dates from both tables in ascending order.

⇒ sales - date.

⇒ SELECT sale - date

FROM online sales

UNION

SELECT sale - date

FROM store sales

ORDER BY sale - date ASC;

sale - date
2025-01-12
2025-01-20
2025-02-05
2025-02-08
2025-03-10
2025-03-25
2025-04-15
2025-04-18
2025-05-02
2025-05-05

4. All sales date (including duplicates)

List all sales date (with duplicates) using UNION ALL

⇒ sales - date.

→ SELECT sale-date FROM online-sales
UNION ALL

SELECT sale-date FROM both-store-sales.

→ OUTPUT SAME AS IN "2".

5. High-value customer

Combine both tables and list unique customer who made purchases greater than 250.

→ customer-name, amount.

→ SELECT customer-name, amount
FROM online-sales

WHERE amount > 250

UNION

SELECT customer-name, amount

FROM online-sales

WHERE amount > 250;

customer-name	amount
Carol	300
George	310
Henry	270

6. Combine sales data

Combine all records from both tables using union all.

→ customer-name, amount, sale-date.

⇒ SELECT customer-name, amount, sale-date
FROM online-sales

UNION ALL

SELECT customer-name, amount, sale-date
FROM store-sales;

customer-name	amount	sale-date
Alice	150	2025-01-12
Brian	250	2025-02-05
Carol	300	2025-03-10
Daniel	220	2025-04-15
Emma	180	2025-05-02
Fiona	200	2025-01-20
Brian	250	2025-02-08
George	310	2025-03-25
Alice	150	2025-04-18
Henry	270	2025-05-05

7. Add sales Source (label)

Combine both tables but include a new column 'source'
that indicates whether the sale was made online or
store.

⇒ customer-name, amount, sale-date, source.

⇒ SELECT customer-name, amount, sale-date,
'Online' AS source
FROM online-sales
UNION ALL

SELECT customer-name, amount, sale-date, 'store'
AS source
FROM store-sales;

customer-name	amount	sale-date	source
Alice	150	2025-01-12	Online
Brian	250	2025-07-05	Online
Carol	300	2025-03-10	Online
Daniel	220	2025-04-15	Online
Emma	180	2025-05-02	Online
Fiona	200	2025-01-20	Store
Brian	250	2025-02-08	Store
George	310	2025-03-25	Store
Alice	150	2025-04-18	Store
Henry	270	2025-05-05	Store

8. Customers Appearing in Both Tables

Find all customers who appear in both online-sales and store-sales (use UNION ALL, GROUP BY and HAVING)

→ customer-name, occurrences.

⇒ SELECT customer-name,

COUNT(*) AS occurrences

FROM (

SELECT customer-name FROM online-sales

UNION ALL

SELECT customer-name FROM online-store

) AS combined GROUPED BY customer-name

HAVING COUNT(*) > 1;

Customer-name	occurrences
Alice	2
Brian	2

9. Total combined sales

Combine both tables using `union all` and calculate the total sales amount across both $\Rightarrow$ total-amount.

$\Rightarrow$ `SELECT sum(amount) AS total-amount`

`FROM (`

`SELECT amount FROM online-sales`

`UNION ALL`

`SELECT amount FROM store-sales`

`)`

`AS total;`

Total-amount
2300

Bonus:

Create a single query that lists each customer's total combined amount from both tables (sum of all their purchases). $\Rightarrow$ customer-name, total-spent

$\Rightarrow$ `SELECT customer-name, sum(amount) AS total-spent`
`FROM (`

`SELECT customer-name, amount FROM online-store`

`UNION ALL`

`SELECT customer-name, amount FROM store-sales`

`) AS combined`

GROUPED BY customer - name ;

customer - name	total - spent
Alia	300
Brian	500
Carol	300
Daniel	220
Emma	180
Fiona	200
George	310
Henry	270